

THE PHASE DIAGRAM OF CUBICAL SITES: CONNECTIONS, REVERSALS, AND THE SEPARATION OF THE TYPE AND TEST MODEL STRUCTURES

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ABSTRACT. Each variety of bounded-distributive-lattice-based algebras determines a cube category and hence a presheaf site carrying two homotopy theories: the cubical-type model structure of cubical type theory and Cisinski’s test structure. A recent series of papers separated the two on the De Morgan site by an explicit map and an explicit finite object. This paper determines the shape of the comparison as a function of the *operations* of the variety. First, on every cubical site with connections but without reversal (the Dedekind family), all isotropy quotients $\square^n/H$ are contractible in both structures — a single equivariant connection contraction — so no object-level isotropy witness can exist there; the known cartesian witness $\square^2/\Sigma_2$ owes its existence precisely to the absence of connections. Second, on each of the three standard sites with reversal — De Morgan, Kleene, and Boolean — the median-collage separation transfers uniformly: the separating map and the finite test-contractible, non-type-contractible object exist on all three. The transfer mechanism is universal-algebraic: all term identities used by the De Morgan proofs hold automatically in the subvarieties, and the only site-specific inputs are two three-line combinatorial lemmas whose De Morgan witness points are unmixed, hence survive in the free Kleene algebras; the Boolean case is simpler still. Third, on the Boolean site the vertex-anchored equivariant contractions collapse to a single self-dual formula — the multiplexer commutes with complement — and the tiny free algebras make the full Boolean isotropy landscape finitely computable; the fate of the twenty De Morgan-separating subgroups of B_3 on the Boolean site is posed as the natural sequel. All finite claims are machine-verified.

1. INTRODUCTION

Cubical type theories interpret dependent type theory in presheaves over a *cube category*, and the choice of cube category is the choice of an algebraic theory of the interval: faces and degeneracies only (the cartesian site), binary connections $\wedge, \vee$ (the Dedekind site), a De Morgan involution (the De Morgan site of CCHM type theory [6]), the Kleene inequality, or full Boolean complementation. Each such site carries two canonical homotopy theories:

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the *cubical-type model structure*, whose fibrations are the uniform Kan fibrations of the type-theoretic interpretation [6, 9, 12], and the *test model structure* of Cisinski [7], which presents the homotopy theory of spaces. On every site considered here the containment

$$(U) \quad W_{\text{type}} \subseteq W_{\text{test}}$$

holds [14, 13]. Whether it is an equality — whether cubical type theory *presents spaces* — depends on the site, and the question has driven a substantial body of work: the folklore argument of Sattler and Coquand for sites with reversal (2018), the equivariant repair of the cartesian site by Awodey, Cavallo, Coquand, Riehl and Sattler [1], the elimination of reversals by Cavallo and Sattler [5], and, on the De Morgan site, an explicit separating map [13], a complete isotropy classification in dimensions ≤ 3 [8], and an explicit finite object on which the two structures disagree about contractibility [13].

This paper organizes these phenomena into a single statement: *which operations of the interval theory produce separation, and which produce agreement*. The answer is a phase diagram (Figure 1).

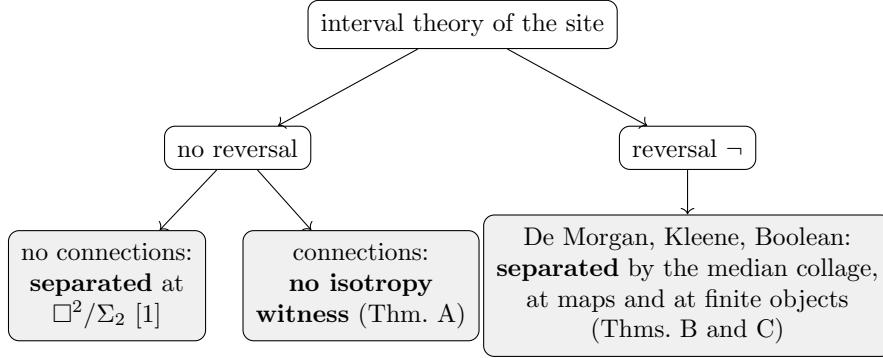


FIGURE 1. The phase diagram of cubical sites.

Theorem 1.1 (Connections kill isotropy witnesses). *Let $\square_{\mathcal{V}}$ be a cube category whose interval theory contains a connection ($\vee$ or $\wedge$) and no reversal, so that the automorphisms of $[n]$ in $\square_{\mathcal{V}}$ form the symmetric group Σ_n . Then for every n and every $H \leq \Sigma_n$ the isotropy quotient $\square^n/H$ admits a strict H -equivariant contraction, and is therefore contractible in both the type and the test structure. In particular the two structures agree on all isotropy quotients of $\square_{\mathcal{V}}$, and no analogue of the known object-level witnesses exists on such sites.*

Theorem 1.2 (Uniform map-level separation). *On each of the De Morgan, Kleene, and Boolean cubical sites, the median collage $\bar{\Phi} : J/K \rightarrow \square^2/K$ of [13] is a test weak equivalence that is not a type weak equivalence. Hence $W_{\text{type}} \subsetneq W_{\text{test}}$ on all three sites.*

Theorem 1.3 (Uniform object-level separation). *On each of the three sites of Theorem 1.2, the subdivided mapping cone C of [13] is an explicit finite presheaf that is test-contractible but not type-contractible: the two structures do not agree about which finite objects are aspherical.*

Proposition 1.4 (Vertex-anchored contraction, Boolean form). *On the Boolean site, for every n and every subgroup $H \leq B_n$ of the signed permutation group that fixes a vertex of $\square^n$, the single formula $c_t(x)_i = (x_i \wedge \neg t) \vee (v_i \wedge t)$ is an H -equivariant strict contraction of $\square^n/H$; the quotient is contractible in both structures. On the De Morgan site the same conclusion holds, but not by this formula: the multiplexer does not commute with the De Morgan involution, and the equivariant contraction must be assembled from the dual pair $x \wedge \neg t$, $x \vee t$ as in [8].*

The proofs of Theorems 1.2 and 1.3 are not re-runs of the De Morgan arguments but an *audit* of them, and the audit is where the interest lies. The separation proofs of [13] decompose into two strata:

- *site-uniform machinery* — uniform-fibration descent, the algebraic fibrant replacement and its coequalizer presentation, strictification of maps out of strict quotients, the equivariant filler slide, the equivariant descent dictionary, the sector/first-exit induction — none of which uses anything about the free algebras beyond the presence of $\wedge, \vee, \neg$ and diagonals; and
- *site-specific combinatorics* — exactly two three-line parity lemmas about free De Morgan algebras on two generators, plus the non-existence of self-negated elements.

For the first stratum, no verification at all is needed on the new sites: every term identity used (self-duality of the median interpolation, the flow identities, the equivariance identities of the collage) is an identity of the De Morgan variety, and Kleene and Boolean algebras form *subvarieties*, so the identities hold there automatically. The transfer of the second stratum is a small miracle of the combinatorics: the two De Morgan parity lemmas are proved by evaluating at two special points of the literal cube, and both points are *unmixed* — they survive in the Berman–Mukaidono representation of free Kleene algebras as monotone functions on the unmixed poset [2] — so the proofs transfer verbatim; on the Boolean site, where negation is output complementation, the lemmas are one-line evaluations at a diagonal point. The resulting picture is summarized by the slogan:

the homotopy-theoretic comparison is controlled by universal algebra: subvariety inheritance transfers the machinery for free, and the whole site-dependence is concentrated in two evaluations at unmixed points.

Two comments on the shape of the diagram. First, it is not monotone in the strength of the theory: the cartesian site (fewest operations) separates, the Dedekind family (add connections) has no isotropy witnesses, and the

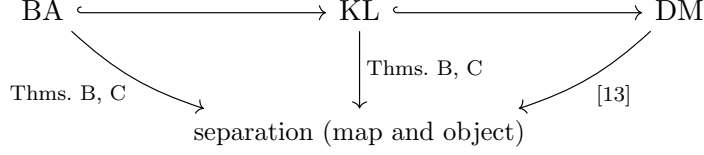


FIGURE 2. Subvariety inclusions $BA \subset KL \subset DM$ and the direction of transfer: identities descend to subvarieties automatically; the two parity lemmas are re-proved at unmixed points.

reversal sites (add $\neg$, up to full Boolean strength) separate again. Adding operations first *heals* the comparison and then *breaks* it; and within the broken phase, further operations simplify the agreement witnesses (Proposition 1.4) without closing the separation. Second, the diagram explains the resistance of the well-known open case: whether the Dedekind site presents spaces cannot be decided by isotropy witnesses — by Theorem 1.1 there are none — which is consistent with the fact that this is precisely the case that remains open.

Relation to the literature. Theorem 1.1 is elementary once stated, but we have not found it stated; it sharpens the observation of Coquand (for $\square^2/\Sigma_2$ with connections) to all subgroups in all dimensions, uniformly in the site. Theorems 1.2 and 1.3 extend the main results of [13] from the De Morgan site to the Kleene and Boolean sites; the Kleene case answers a question left open by the reversal-quotient analysis of [14], which treated the Kleene interval quotient but not the separation. Cavallo and Sattler [5] show that reversals can be eliminated from cubical *type theories*; our results complement theirs on the model-structure side: every standard site whose interval theory contains a reversal fails to present spaces, and does so by the same explicit witness.

Machine verification. All finite combinatorial claims — the stage-zero censuses on the Kleene and Boolean sites, the parity lemmas, the stratification checks, and the negative controls — are verified by exhaustive computation; the scripts (`kleene_site.py`, `boolean_site.py`) are archived with the artifact of the series (Section 6).

2. CUBICAL SITES FROM VARIETIES

2.1. The sites. Let $\mathcal{V}$ be a variety of algebras on the signature $\{\wedge, \vee, 0, 1\}$ or $\{\wedge, \vee, \neg, 0, 1\}$ containing the bounded distributive lattice axioms, and write $F_{\mathcal{V}}(n)$ for the free algebra on $x_1, \dots, x_n$. The *cube category* $\square_{\mathcal{V}}$ has objects $[n]$ for $n \geq 0$ and morphisms

$$\square_{\mathcal{V}}([m], [n]) = \text{Hom}_{\mathcal{V}}(F_{\mathcal{V}}(n), F_{\mathcal{V}}(m)) \cong F_{\mathcal{V}}(m)^n,$$

composition being substitution; presheaves on $\square_{\mathcal{V}}$ are $\mathcal{V}$ -cubical sets. We consider:

- the *Dedekind family*: $\mathcal{V} \supseteq$ bounded distributive lattices, no $\neg$ (this includes the Dedekind site itself, $F_{\mathcal{V}}(n)$ the free bounded distributive lattice, of cardinality the Dedekind numbers 3, 6, 20, 168, $\dots$);
- the *De Morgan site* $\square_{\text{DM}}$: $\mathcal{V} =$ De Morgan algebras (distributive lattices with an involution $\neg$ satisfying the De Morgan laws); $|F_{\text{DM}}(2)| = 168$;
- the *Kleene site* $\square_{\text{KL}}$: the subvariety of De Morgan algebras with $x \wedge \neg x \leq y \vee \neg y$; $|F_{\text{KL}}(n)| = 6, 84, 43918, \dots$ [2];
- the *Boolean site* $\square_{\text{BA}}$: the subvariety of Boolean algebras; $|F_{\text{BA}}(n)| = 2^{2^n}$;
- the *cartesian site* $\square_{\text{cart}}$: no lattice operations; morphisms are tuples of variables and constants.

All of these are strict test categories: the cube category with connections is a strict test category by Maltsiniotis [11], and the remaining cases follow from the criteria of [7]; see the systematic treatment of Buchholtz and Morehouse [4]. Hence each site carries the test model structure, with $\mathbf{W}_{\text{test}}$ created by the nerve of the category of elements $\text{el}(-)$, and each carries the cubical-type model structure with cofibrations the monomorphisms and fibrations the uniform Kan fibrations generated by the open-box inclusions $(m \hat{\otimes} \delta^e)$ [9, 12]. The containment (U) holds on every strict test cube category by the two-out-of-three argument of [13, Prop. 2.5] (which we do not repeat): the generating trivial cofibrations are monic test equivalences, and type-trivial fibrations have the right lifting property against all monomorphisms.

2.2. Symmetries and isotropy quotients. The invertible morphisms $[n] \rightarrow [n]$ of $\square_{\mathcal{V}}$ form the symmetric group Σ_n when $\neg \notin \mathcal{V}$ and the signed permutation (hyperoctahedral) group $B_n = \Sigma_n \ltimes (\mathbb{Z}/2)^n$ when $\neg \in \mathcal{V}$. For H a subgroup of this automorphism group, the *isotropy quotient* $\square^n/H$ is the levelwise orbit presheaf. These quotients are the natural finite testing ground for the comparison: they are where all known object-level phenomena live [14, 8, 13].

2.3. Free-algebra representations. Computations use the standard function representations. Free De Morgan algebras are monotone $\{0, 1\}$ -valued functions on the literal cube $L_n = \{0, 1\}^{2^n}$ (coordinates in pairs $(x_i, \neg x_i)$), with negation $(\neg\varphi)(p) = 1 - \varphi(\rho p)$, where ρ swaps and complements each pair. Free Kleene algebras are the monotone functions on the *unmixed* subposet $U_n \subseteq L_n$ — the points having no $(0, 0)$ -pair together with a $(1, 1)$ -pair — with the same formula for negation (ρ preserves U_n) [2]. Free Boolean algebras are all $\{0, 1\}$ -valued functions on $\{0, 1\}^n$, with negation the output complement. A point of L_n is called unmixed when it lies in U_n ; the transfer of the parity lemmas in Section 4 rests on the observation that the De Morgan proofs only ever evaluate at unmixed points.

3. CONNECTIONS KILL ISOTROPY WITNESSES

Proof of Theorem 1.1. Suppose $\vee \in \mathcal{V}$ (the case of $\wedge$ is dual, using the 0-endpoint). Consider the cell

$$c = (x_1 \vee t, \dots, x_n \vee t) \in \square_{\mathcal{V}}([n+1], [n]),$$

regarded as a map $c : \square^n \otimes \square^1 \rightarrow \square^n$ of representables, where t is the cylinder coordinate. Then: (i) $c|_{t=0} = \text{id}$ and $c|_{t=1}$ is the constant map at the vertex $(1, \dots, 1)$; (ii) c is Σ_n -equivariant: permuting the x_i commutes with the coordinatewise formula, and the vertex $(1, \dots, 1)$ is fixed; (iii) since quotients of presheaves are levelwise and the cylinder is formed in the x -coordinates only, c descends to a homotopy

$$\bar{c} : (\square^n/H) \times \square^1 \longrightarrow \square^n/H, \quad \bar{c}_0 = \text{id}, \quad \bar{c}_1 = \text{const},$$

for every $H \leq \Sigma_n$. A strict homotopy between the identity and a constant is a left homotopy in any model structure in which the projection $X \times \square^1 \rightarrow X$ is a weak equivalence, which holds in the type structure (the interval is contractible) and in the test structure (the site is a strict test category). Hence $\square^n/H \rightarrow 1$ is a weak equivalence in both structures. $\square$

Corollary 3.1. *On the Dedekind site the type and test structures agree on all isotropy quotients. In particular, whether the Dedekind site presents spaces cannot be decided by an isotropy witness.*

Remark 3.2. On the cartesian site the contraction of Theorem 1.1 is unavailable — there is no cell $x \vee t$ — and indeed the symmetric square $Q = \square^2/\Sigma_2$ is a witness: it is test-contractible, while its failure to be type-contractible is the observation of Coquand and Sattler that motivated the equivariant model structure of [1]. The phase boundary between the cartesian and Dedekind columns of Figure 1 is thus exactly the presence of a connection.

4. THE UNIFORM SEPARATION

We recall the shape of the De Morgan separation [13]. $K = \{1, \text{sw}, \text{n}, g\}$ is the Klein group acting on the square by the coordinate swap sw , the double reversal $\text{n}(x, y) = (\neg x, \neg y)$, and $g = \text{nsw}$; $W := \square^2/K$. The *join model* J is the double mapping cylinder of the two strata circles, with a twisted K -action, and the *median collage*

$$\Phi = (F(d, a, t), F(d, \neg a, t)), \quad F(d, a, t) := (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a),$$

descends to $\bar{\Phi} : J/K \rightarrow W$. The main theorems of [13] state that $\bar{\Phi}$ is a test weak equivalence (both sides realize $B\mathbb{Z}/2 \star B\mathbb{Z}/2$) but not a type weak equivalence (the *nullity theorem*: every map from W into a fibrant replacement of J/K is null-homotopic), and that a subdivided mapping cone C of $\bar{\Phi}$ is test-contractible but not type-contractible. All constructions make sense verbatim on the Kleene and Boolean sites: the formulas for the actions, the join model, and F use only $\wedge, \vee, \neg, 0, 1$.

4.1. The audit.

Lemma 4.1 (Subvariety inheritance). *Every term identity of the De Morgan variety holds in the Kleene and Boolean varieties. In particular the self-duality $\neg F(d, a, t) = F(\neg d, \neg a, t)$, the flow identities $F(d, a, t \wedge s) = F(d, F(d, a, t), s)$ and $F(d, a, t \vee s) = F(F(d, a, t), a, s)$, the equivariance identities of the collage, and the sector formulas of the subdivided cone hold on all three sites.*

Proof. Kleene and Boolean algebras are subvarieties of De Morgan algebras: each is axiomatized by the De Morgan axioms plus further identities/inequalities. Identities valid in a variety hold in every subvariety. $\square$

Lemma 4.2 (No self-negated elements, all three sites). *On each site, no element of a free algebra equals its own negation. Consequently the double reversal n acts freely on the nonempty cells of the square, at every level.*

Proof. De Morgan and Kleene: the all- $(1, 0)$ point p_0 of the literal cube (each pair equal to $(1, 0)$) satisfies $\rho p_0 = p_0$ and is unmixed, hence lies in U_n ; a self-negated φ would satisfy $\varphi(p_0) = 1 - \varphi(p_0)$. Boolean: negation is the output complement, so $\neg \varphi \neq \varphi$ pointwise. $\square$

Lemma 4.3 (First parity lemma, all three sites). *No φ in the free algebra on two generators satisfies $\varphi^{\text{sw}} = \neg \varphi$.*

Proof. De Morgan and Kleene: the point $q = (1, 0, 1, 0)$ satisfies $\text{sw } q = q$ and $\rho q = q$, and is unmixed, hence lies in U_2 ; at q the condition reads $\varphi(q) = 1 - \varphi(q)$. Boolean: at a diagonal input (a, a) the condition reads $\varphi(a, a) = 1 - \varphi(a, a)$. $\square$

Lemma 4.4 (Second parity lemma, all three sites). *No φ in the free algebra on two generators satisfies both $\varphi^{\text{sw}} = \varphi$ and $\varphi^n = \neg \varphi$.*

Proof. De Morgan and Kleene: the point $m = (1, 0, 0, 1)$ is unmixed and its bitwise complement equals $\text{sw } m$; symmetry gives $\varphi(\overline{m}) = \varphi(m)$ while the second condition gives $\varphi(\overline{m}) = 1 - \varphi(m)$. Boolean: at the input $(0, 1)$, the n -condition reads $\varphi(1, 0) = 1 - \varphi(0, 1)$ while symmetry gives $\varphi(1, 0) = \varphi(0, 1)$. $\square$

Proof of Theorems 1.2 and 1.3. We audit the proofs of the corresponding theorems of [13], replacing the De Morgan site by the Kleene or Boolean site throughout, and verify that every ingredient remains valid.

Site-uniform machinery. The following results of [13] are proved for an arbitrary presheaf category over a strict test cube category whose interval theory contains $\wedge, \vee, \neg$ and diagonals, and use no further structure of the free algebras: the descent lemma for uniform fillers and the algebraic fibrant replacement with its coequalizer presentation (the proofs use cellular decompositions of monomorphisms, pullback of boxes to representables, and strict naturality — all site-independent); the strictification of maps out of strict quotients (Yoneda plus levelwise colimits); the equivariant filler slide (the

slide formulas $\tau \wedge \neg z$, $\tau \vee z$ exist by Lemma 4.1); the upper bound (U); the equivariant descent dictionary and the sector, pasting, and first-exit lemmas of the object-level proof (coequalizer relations plus diagonals, which all our sites possess); and the collage formula for nerves of sieve-cosieve decompositions (general category theory [3, 15]).

Term identities. Every algebraic identity invoked — self-duality and flow for F , the equivariance of Φ and of the cone slide, the face computations — holds by Lemma 4.1.

The test side. The stratification of the deck action on the cells of the square is site-uniform once Lemma 4.2 is in place: $\mathbf{sw}$ -fixed cells are the diagonal ones (φ, φ) , g -fixed cells the antidiagonal ones $(\varphi, \neg\varphi)$, and $\mathbf{n}$ acts freely; the strata are copies of the interval quotient $\square^1/\neg$, whose category of elements is $B\mathbb{Z}/2$ by the free-covering argument (freeness is again Lemma 4.2); the free parts are contractible covers with the generic cells as terminal objects (stabilizer-freeness of the generic cell: $\mathbf{sw}$ and g move it since $y \neq x$ and $\neg y \neq x$ in the free algebra on two generators, and $\mathbf{n}$ by Lemma 4.2). The collage computation then gives $\mathrm{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$ on each site, and the three-legs argument for $\mathrm{el}(\bar{\Phi})$ being a weak equivalence proceeds verbatim: the collapse-locus closure and flow retraction use only the flow identities and Lemma 4.2; the middle layers use the terminal covers and free quotients. We have verified the stratification and the census ingredients by machine on both new sites (Section 6).

The type side. The nullity theorem’s stage induction is site-uniform; its base is the rigidity of invariant cylinder classes of $(J/K)([2])$, which is a formal consequence of Lemmas 4.3 and 4.4 exactly as in [13]: invariance under $\mathbf{sw}$ modulo the deck group forces componentwise $\mathbf{sw}$ -symmetry by Lemma 4.3, and then invariance under $\mathbf{n}$ forces componentwise strict invariance by Lemma 4.4. The machine censuses confirm the rigidity exhaustively: on the Kleene site $(J/K)([2])$ has 96 invariant interior classes, every one with a strictly invariant representative; on the Boolean site, 8, likewise; and on neither site does any invariant class map to the identity class of W under the collage (no strict sections; on the Boolean site 64 non-invariant triples do hit the identity class — a genuinely Boolean phenomenon — but none is source-invariant). Non-contractibility of W in the type structure follows from (U) and $\mathrm{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$ as in [13].

Assembly. With all ingredients in place, the proofs of the main theorems of [13] go through verbatim on each site: $\bar{\Phi}$ is a test weak equivalence but not a type weak equivalence (Theorem 1.2), and the subdivided mapping cone is test-contractible but not type-contractible (Theorem 1.3). $\square$

Remark 4.5 (What does *not* transfer). The transfer is not a triviality about quotients of proofs: the stage-zero *numerics* differ substantially across the sites (Table 1), and at least one qualitative phenomenon is new on the Boolean site — the existence of (non-invariant) collage preimages of the identity class, which on the De Morgan site is excluded outright. What transfers

is the *mechanism*, because its site-specific content was already concentrated in Lemmas 4.3–4.4 by [13].

	De Morgan	Kleene	Boolean
$ F(2) $	168	84	16
classes of $W([2])$	7140	1806	72
strict endomorphism classes	72	38	12
invariant interior classes of $(J/K)([2])$	360	96	8
of which essential	0	0	0
collage-image = id triples	0	0	64
of which source-invariant	0	0	0

TABLE 1. Stage-zero censuses across the three reversal sites (machine-verified; De Morgan values from [13, 14]).

5. VERTEX-ANCHORED CONTRACTIONS ON THE BOOLEAN SITE

On the separation side, the Boolean site behaves like the De Morgan site. On the agreement side, the extra Boolean operations do not (so far as we prove here) enlarge the class of agreeing subgroups, but they simplify its witnesses to a single self-dual formula:

Lemma 5.1 (The multiplexer commutes with complement). *In Boolean algebra, $\neg((x \wedge \neg t) \vee (v \wedge t)) = (\neg x \wedge \neg t) \vee (\neg v \wedge t)$.*

Proof. Expand: the left side is $(\neg x \vee t) \wedge (\neg v \vee \neg t) = (\neg x \wedge \neg v) \vee (\neg x \wedge \neg t) \vee (t \wedge \neg v)$, using $t \wedge \neg t = 0$; and $\neg x \wedge \neg v = \neg x \wedge \neg v \wedge (t \vee \neg t)$ is absorbed by the last two joinands. $\square$

Note where this computation fails on the De Morgan site: $t \wedge \neg t \neq 0$ there, and the leftover joinand is exactly the obstruction. The conclusion of Proposition 1.4 survives on the De Morgan site by a different route — an equivariant interpolation assembled from the dual pair $x \wedge \neg t$, $x \vee t$, with equivariance via De Morgan duality [8] — but the single complement-equivariant formula does not exist there.

Proof of Proposition 1.4. Let $v = (v_1, \dots, v_n) \in \{0, 1\}^n$ be a vertex fixed by $H \leq B_n$, and set

$$c_t(x)_i = (x_i \wedge \neg t) \vee (v_i \wedge t),$$

a cell of the Boolean site. For $h = (\sigma, s) \in H$ acting by $h(x)_i = \neg^{s_i} x_{\sigma(i)}$:

$$h(c_t(x))_i = \neg^{s_i} ((x_{\sigma(i)} \wedge \neg t) \vee (v_{\sigma(i)} \wedge t)) = (\neg^{s_i} x_{\sigma(i)} \wedge \neg t) \vee (\neg^{s_i} v_{\sigma(i)} \wedge t) = c_t(h(x))_i,$$

the middle equality by Lemma 5.1 and the last because h fixes v . So c is H -equivariant, descends to the quotient, and interpolates strictly between the identity and the constant at the $(H$ -fixed) image of v ; conclude as in Theorem 1.1. $\square$

Remark 5.2. A subgroup fixing a vertex always fixes a cell (the appropriately signed diagonal), so on any of our sites it belongs to the fixed-cell family, which agrees [8]. The genuinely open Boolean question concerns the *other* side of the De Morgan classification: of the twenty subgroups of B_3 whose De Morgan quotients are separated [8], none fixes a vertex, and their Boolean quotients are not decided by anything in this paper — the De Morgan separation certificates rest on parity arguments in $F_{\text{DM}}(3)$ that have no direct Boolean counterpart, while the Boolean free algebras are so small ($|F_{\text{BA}}(3)| = 256$) that the complete Boolean isotropy classification in dimension three is a finite computation. We pose it as the natural sequel.

6. MACHINE CROSS-CHECKS

The scripts `kleene_site.py` and `boolean_site.py`, archived in the artifact repository of this series,

`https://github.com/r-shibusawa/cubical-strict-j`

(release v1.17.0, commit 6365ec7, archived at DOI 10.5281/zenodo.21987530), verify exhaustively: the Berman–Mukaidono representation ($|U_2| = 14$, $|F_{\text{KL}}(1)| = 6$, $|F_{\text{KL}}(2)| = 84$, closure under $\rho, \text{sw}, \text{n}$ and substitution); Lemmas 4.2–4.4 by direct enumeration on both new sites; the deck stratification at levels ≤ 3 (Boolean) and ≤ 2 (Kleene); the rigidity censuses and the absence of essential invariant classes (Table 1); the absence of source-invariant collage sections; and, on the Boolean site, the strict-homotopy component of the identity among the 12 strict endomorphism classes (it is $\{\text{id}\}$, and contains no constant, the stage-zero shadow of Theorem 1.2). Lemma 5.1 and the equivariance computation of Theorem 1.4 are verified over the generating algebras of the varieties.

7. OUTLOOK

The phase diagram invites a finer parametrization. Between the Dedekind and De Morgan columns sit sites with a reversal but weaker lattice laws, and beyond Boolean nothing remains; within the reversal phase, Theorem 1.4 shows the agreement region grows with the strength of the theory while the separation persists. Three directions seem worthwhile. First, the Boolean isotropy classification in dimensions ≤ 3 : with $|F_{\text{BA}}(3)| = 256$ every census is small, and the mux contraction settles all vertex-fixing subgroups, so only the fixed-point-free mixed subgroups remain — a complete Boolean analogue of [8] is within reach. Second, the Dedekind question itself: Theorem 1.1 rules out isotropy witnesses, so a separation, if it exists, must be found among maps or non-quotient objects, while a proof of agreement would need the localizer-generation methods that the present series has so far only calibrated. Third, the variety-indexed point of view suggests a general conjecture: for a cube category arising from a variety $\mathcal{V}$ of De Morgan-type algebras, the type and test structures on $\mathcal{V}$ -cubical sets coincide if and only if $\mathcal{V}$ has no reversal and a connection — the two phases of Figure 1 that

our witnesses cannot reach are exactly the two for which no separation is known.

Data availability. The verification scripts are openly available in the repository named in Section 6, archived on Zenodo: release `v1.17.0`, commit `6365ec7`, DOI [10.5281/zenodo.21987530](https://doi.org/10.5281/zenodo.21987530).

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